

AP CALCULUS AB - UNIT 1

1.11 Defining Continuity at a Point

Lesson Overview

- State and apply the three conditions for continuity at a point.
- Identify the first failed condition from graphs, tables, formulas, and piecewise rules.
- Use continuity of standard functions and valid compositions.
- Reject common invalid converses about limits and point values.

Key Knowledge and Vocabulary

Condition 1. $f(c)$ is defined.

Condition 2. $\lim_{x \rightarrow c} f(x)$ exists as a finite number.

Condition 3. $\lim_{x \rightarrow c} f(x) = f(c)$.

First failed condition. Report the earliest precise failure rather than saying only “the graph is broken.”

Local property. Continuity at c depends only on the function near c and at c .

Explanation and Strategy

Step 1. Audit in order: domain, two-sided limit, equality.

Step 2. For a formula built from standard continuous functions, check the domain and then substitute.

Step 3. A corner may be continuous; differentiability is not required for continuity.

Solved Examples

Worked Example: All conditions hold

Verify continuity of $f(x) = \sqrt{x+5}$ at $x = 4$.

Solution. $f(4) = 3$ exists. The square-root function is continuous on its domain, so $\lim_{x \rightarrow 4} f(x) = 3 = f(4)$. All three conditions hold.

Worked Example: Wrong point value

Let $g(x) = x + 2$ for $x \neq 1$ and $g(1) = 10$. Is g continuous at 1?

Solution. $g(1)$ exists, and the limit is 3, but $3 \neq 10$. Condition 3 fails.

Worked Example: Piecewise join

$h(x) = 2x + 1$ for $x < 3$ and $h(x) = x + 4$ for $x \geq 3$. Verify continuity at 3.

Solution. Left limit 7, right limit 7, and $h(3) = 7$. Thus h is continuous at 3.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. List the three continuity conditions at $x = c$.

2. Is $f(x) = x^2 + 1$ continuous at $x = -3$?

3. Is $g(x) = 1/(x - 2)$ continuous at $x = 2$?

4. Can a function with a corner be continuous?

Medium Level

Apply the idea in one or two connected steps.

5. For $f(x) = |x - 1|$, verify continuity at $x = 1$.

6. A table approaches 4 from both sides at $x = 2$, and the center row is $f(2) = 4$. What does the table support?

7. A graph approaches 6 from both sides but has no point at $x = 3$. Which condition fails first?

8. A graph has $f(0) = 2$, left limit 2, and right limit 5. Which condition fails first?

Hard Level

Connect representations, conditions, and justification.

9. Determine whether $f(x) = \frac{x^2-9}{x-3}$ is continuous at 3 as originally defined.

10. Find k so that $f(x) = \begin{cases} x^2 + k, & x < 2 \\ 3x - 1, & x \geq 2 \end{cases}$ is continuous at 2.

11. Is $f(x) = \sqrt{4-x^2}$ continuous at $x = 2$ relative to its domain?

12. Explain why $\lim_{x \rightarrow c} f(x) = f(c)$ automatically includes the existence of both quantities when written as a valid equality.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Find all a for which $f(x) = \frac{x^2-a^2}{x-a}$ with the original quotient formula is continuous at $x = a$.

14. Suppose f and g are continuous at c , but $g(c) = 0$. Is f/g continuous at c ?

15. Give a counterexample to: “If $f(c)$ exists, then f is continuous at c .”

Challenge Level

Synthesize, construct, generalize, or prove.

16. Prove that if f is continuous at c and $f(c) > 0$, then $f(x) > 0$ for all x sufficiently close to c .

17. Construct a function continuous at 0 but discontinuous at every point $1/n$ for positive integers n .

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. Which condition is not required for continuity at c ?

- | | |
|--|--|
| (A) $f(c)$ exists | (B) $\lim_{x \rightarrow c} f(x)$ exists |
| (C) $\lim_{x \rightarrow c} f(x) = f(c)$ | (D) $f'(c)$ exists |

2. A function has $f(2) = 3$ and $\lim_{x \rightarrow 2} f(x) = 3$. Then f is

- | | |
|-------------------------------------|------------------------|
| (A) continuous at 2 | (B) discontinuous at 2 |
| (C) necessarily differentiable at 2 | (D) undefined at 2 |

3. If the one-sided limits at c are unequal, which continuity condition fails first?

- | | |
|-------------------|---------------------------------|
| (A) Point value | (B) Two-sided limit existence |
| (C) Equality only | (D) Domain of every other point |

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator graph of $f(x) = |x|$ has a sharp corner at 0. Which statement is correct?

- (A) f is not continuous at 0
(B) f is continuous at 0 but not differentiable there
(C) $f(0)$ is undefined
(D) The limit is infinite

5. A numerical table approaches 2.000 from both sides and reports $f(c) = 2.000$. What is the most accurate claim?

- (A) Continuity is proved.
(B) The data strongly support continuity, but exact proof may require formula or theorem evidence.
(C) The limit DNE.
(D) The function is differentiable.

Free-Response Practice – No Calculator

$$\text{Let } f(x) = \begin{cases} ax + 2, & x < 1 \\ x^2 + a, & x \geq 1 \end{cases}.$$

(a) Write the left-hand limit and the right-hand limit/value at $x = 1$. [2 points]

(b) Is there a value of a that makes f continuous at 1? [2 points]

(c) Identify the continuity condition that fails for every a . [2 points]

Free-Response Practice – Calculator Required

A calculator table for g near $x = 4$ approaches 6.000 from both sides, while the function screen reports $g(4) = 6$.

(a) State the three continuity conditions and how the evidence addresses each. [3 points]

(b) What additional evidence would convert the numerical support into an exact proof? [2 points]

(c) Why does the table not establish differentiability?

[1 points]